

Data Analytics Course Syllabus

Archer Infotech, Kothrud Pune | Classroom & live-online batches | Updated September 2026

1. Introduction to Data Analytics

1.1 Data Analytics Fundamentals

- What is Data Analytics?
- Importance of Data Analytics
- Data-Driven Decision Making
- Role and Responsibilities of a Data Analyst
- Data Analyst vs Data Scientist
- Data Analyst vs Business Analyst
- Data Analyst vs BI Developer

1.2 Types of Data Analytics

- Descriptive Analytics
- Diagnostic Analytics
- Predictive Analytics – Introduction
- Prescriptive Analytics – Introduction

1.3 Data Analytics Lifecycle

- Business Problem Understanding
- Data Collection
- Data Cleaning
- Data Transformation
- Exploratory Data Analysis
- Visualization
- Insight Generation
- Reporting
- Decision Making

1.4 Types of Data

- Structured, Semi-Structured and Unstructured Data
- Numerical and Categorical Data
- Ordinal and Nominal Data
- Time-Series Data

2. Excel for Data Analytics

2.1 Excel Fundamentals

- Workbook, Worksheet, Rows, Columns, Cells
- Relative, Absolute and Mixed Cell References
- Data Types and Formatting

2.2 Core Functions

- SUM, AVERAGE, MIN, MAX
- COUNT, COUNTA, COUNTBLANK
- ROUND, ROUNDUP, ROUNDDOWN

2.3 Logical and Conditional Functions

- IF, Nested IF, IFS
- AND, OR, NOT

- IFERROR, IFNA
- SUMIF, SUMIFS
- COUNTIF, COUNTIFS
- AVERAGEIF, AVERAGEIFS

2.4 Text Functions

- LEFT, RIGHT, MID
- LEN, TRIM, CLEAN
- UPPER, LOWER, PROPER
- CONCAT, TEXTJOIN
- FIND, SEARCH
- SUBSTITUTE, REPLACE

2.5 Date and Time Functions

- TODAY, NOW, DATE
- DAY, MONTH, YEAR
- DATEDIF, EDATE, EOMONTH
- WEEKDAY, NETWORKDAYS, WORKDAY

2.6 Lookup Functions

- VLOOKUP, HLOOKUP
- XLOOKUP, XMATCH
- INDEX, MATCH, INDEX-MATCH

2.7 Data Cleaning in Excel

- Remove Duplicates
- Text to Columns
- Flash Fill
- Find and Replace
- Handling Blank Cells and Invalid Data
- Data Validation
- Standardizing Data

2.8 Sorting, Filtering and Tables

- Single-Level and Multi-Level Sorting
- Auto Filter, Custom Filter, Advanced Filter
- Excel Tables and Structured References
- Total Row

2.9 Conditional Formatting

- Highlight Cell Rules
- Top/Bottom Rules
- Data Bars, Color Scales, Icon Sets
- Formula-Based Conditional Formatting

2.10 Pivot Tables and Pivot Charts

- Rows, Columns, Values and Filters
- Grouping
- Calculated Fields
- Slicers and Timelines
- Pivot Charts

2.11 Excel Charts

- Column, Bar, Line, Pie and Doughnut
- Area, Scatter, Combo, Waterfall

- Histogram and Sparklines

2.12 Advanced Excel

- Named Ranges
- Dynamic Arrays
- FILTER, SORT, SORTBY, UNIQUE, SEQUENCE, LET
- Data Consolidation
- What-If Analysis
- Goal Seek
- Scenario Manager

3. Power Query

- Introduction to Power Query and ETL
- Connecting to Excel, CSV, Text, Folder, Web and Databases
- Data Cleaning and Type Conversion
- Split/Merge Columns
- Conditional and Custom Columns
- Text, Number and Date Transformations
- Append and Merge Queries
- Pivot and Unpivot
- Parameters and Query Dependencies
- Introduction to M Language

4. SQL for Data Analytics

- Database, DBMS, RDBMS, Schema, Tables, Keys and Relationships
- CREATE, INSERT, UPDATE, DELETE, SELECT
- DISTINCT, Aliases, LIMIT/TOP
- WHERE, AND, OR, NOT, IN, BETWEEN, LIKE, NULL
- ORDER BY
- COUNT, SUM, AVG, MIN, MAX
- GROUP BY and HAVING
- String and Date Functions
- CASE Expressions
- INNER, LEFT, RIGHT, FULL, CROSS and SELF JOIN
- Subqueries and Correlated Subqueries
- CTEs and Recursive CTE Introduction
- Window Functions: OVER, PARTITION BY, ROW_NUMBER, RANK, DENSE_RANK, LAG, LEAD
- Running Totals and Moving Averages
- Views
- Sales, Customer, Revenue and Product Analysis
- Month-over-Month and Year-over-Year Analysis

5. Statistics for Data Analytics

- Population, Sample, Parameter and Statistic
- Mean, Median, Mode and Weighted Mean
- Range, Variance, Standard Deviation, IQR
- Percentiles and Quartiles
- Normal Distribution, Skewness and Kurtosis
- Probability and Conditional Probability
- Sampling Methods and Sampling Bias
- Correlation and Covariance

- Pearson and Spearman Correlation
- Hypothesis Testing
- Null and Alternative Hypothesis
- P-Value and Significance Level
- Type I and Type II Errors
- Z-Test, T-Test, Chi-Square Test and ANOVA Introduction

6. Python Fundamentals for Data Analytics

- Python Installation, Jupyter Notebook and Google Colab
- Variables, Data Types and Operators
- Lists, Tuples, Sets and Dictionaries
- if/elif/else
- for and while Loops
- Functions and Lambda Functions
- File Handling: Text, CSV, JSON and Excel
- Exception Handling

7. NumPy for Data Analytics

- NumPy Arrays
- Shape, Dimensions and Data Types
- Indexing, Slicing, Filtering and Reshaping
- Numerical and Aggregate Operations
- Mean, Median, Variance and Standard Deviation
- Percentiles
- Broadcasting and Vectorization
- Random Numbers

8. Pandas for Data Analytics

- Series and DataFrame
- Importing CSV, Excel, JSON and SQL Data
- head(), tail(), sample(), info(), describe()
- loc, iloc and Boolean Filtering
- Missing Values and Duplicate Records
- Data Type and String Cleaning
- isnull(), dropna(), fillna()
- rename(), replace(), astype(), apply(), map()
- Sorting and Ranking
- groupby() and agg()
- concat(), merge(), join()
- pivot(), pivot_table(), melt()
- Date and Time Analysis

9. Data Cleaning and Preparation

- Data Quality Dimensions
- Missing Data
- Duplicate Data
- Invalid and Inconsistent Data
- Data Type Conversion
- Outlier Detection: Box Plot, IQR and Z-Score
- Data Standardization

10. Exploratory Data Analysis

- Dataset Understanding and Profiling
- Univariate Analysis
- Bivariate Analysis
- Multivariate Analysis
- Correlation Analysis
- Pattern and Trend Identification
- Business Insight Generation

11. Data Visualization

- Matplotlib
- Seaborn
- Plotly
- Line, Bar, Histogram, Scatter and Box Plots
- Heatmaps and Pair Plots
- Interactive Visualizations
- Visualization Best Practices

12. Power BI

- Power BI Desktop and Service
- Data Import
- Power Query in Power BI
- Data Modeling and Relationships
- Star Schema
- Measures, Calculated Columns and Calculated Tables
- DAX: SUM, SUMX, COUNTROWS, DISTINCTCOUNT, AVERAGE, IF, SWITCH, DIVIDE
- CALCULATE, FILTER, ALL, ALLEXCEPT, VALUES, SELECTEDVALUE
- Time Intelligence: YTD, MTD, QTD, SAMEPERIODLASTYEAR, DATEADD
- Tables, Matrix, Cards, KPI Cards and Charts
- Slicers, Drill Down, Drill Through and Tooltips
- Bookmarks, Buttons and Navigation
- Publishing, Workspaces, Sharing and Scheduled Refresh

13. Business Analytics

- Sales Analytics
- Customer Analytics
- Marketing Analytics
- Financial Analytics
- HR Analytics
- Operations Analytics

14. Key Performance Indicators

- KPI vs Metric
- Leading and Lagging Indicators
- Revenue, Profit, Margin and Growth
- Conversion, Retention and Churn
- Average Order Value and Customer Acquisition Cost
- KPI Dashboard Design

15. Data Storytelling

- Data vs Information vs Insight
- Understanding the Audience
- Context, Problem, Evidence, Insight, Recommendation and Action
- Executive Reporting
- Presenting Dashboards and Findings

16. AI-Assisted Data Analytics

- AI for Data Analysts
- Prompting for Analytics
- AI-Assisted Excel Formulas
- Natural Language to SQL
- AI-Assisted Python and Pandas
- AI-Assisted EDA
- Chart and KPI Suggestions
- Executive Summary Generation
- Responsible Use, Validation, Privacy and Hallucination Awareness

17. Git and GitHub

- Repository, Clone, Add, Commit, Push and Pull
- Branches
- GitHub Repositories
- README Files
- Data Analytics Portfolio

18. Projects

- Excel Sales Analysis
- SQL Retail Analysis
- Python Data Analysis
- HR Analytics Dashboard
- E-Commerce Analytics
- Financial Performance Dashboard
- Marketing Analytics Dashboard

19. Capstone Project

- Business Problem Understanding
- Data Collection
- Data Cleaning and Transformation
- Excel Analysis
- SQL Analysis
- Python Analysis
- Statistical Analysis
- EDA
- KPI Identification
- Power BI Data Modeling and DAX
- Dashboard Creation
- Business Insight Generation
- Recommendations
- Executive Presentation
- GitHub Documentation

20. Interview Preparation

- Excel
- SQL
- Python
- Statistics
- Power BI
- Business Case Studies
- Dashboard Presentation
- Mock Interviews
- Microsoft Excel
- Power Query
- Power BI
- DAX
- SQL
- PostgreSQL/MySQL concepts
- Python
- NumPy
- Pandas
- Matplotlib
- Seaborn
- Plotly
- Jupyter Notebook
- Google Colab
- Git
- GitHub
- Generative AI / AI coding assistants

Enquire or book a free demo session. Archer Infotech, Flat No. 12, Divyadarshan Housing Society, behind Kothrud Bus Stand Road, Kothrud, Pune 411038. Phone +91 9850 678451. Email info@archerinfotech.in. Web archerinfotech.in.